AWS FinOps Cost Optimization Report

QC-Validated | READ-ONLY Audit | All Numbers Reconciled

AWS Account	730649731674 (Himanshu Gohil)
Report Date	2026-10-04
Audit Period	Oct 2026 MTD (Oct 1–4) + Sep 2026 (full)
Access Level	READ-ONLY — No resources modified
Report Author	Kiro FinOps Agent
QC Status	PASS — Reconciliation difference $0.00 for both months

Cost Snapshot — Both Months Reconcile to $0.00 Net

Sep 2026 Net Cost (CE total)	**~$0.00**	Oct MTD Net Cost (CE total)	**~$0.00**
Sep 2026 Gross Charges	**$0.4342**	Oct MTD Gross Charges	**$0.0328**
Total Resource Objects	**10**	Active Cost Anomalies	**0**
Confirmed Current Savings	**$0.00**	Conditional Savings (if credits expire)	**$2.80/month**

This report was generated in READ-ONLY mode. No AWS resources were created, modified, stopped, or deleted. All remediation requires explicit human approval. All costs are AWS Cost Explorer UnblendedCost values. Both reporting periods reconcile to $0.00 difference.

AWS FinOps Cost Optimization Report

Generated: *2026-10-04*
AWS Account: *730649731674 (Himanshu Gohil)*
Report Author: *Kiro FinOps Agent*
Access Level: *READ-ONLY No AWS resources were modified*
Audit Period: *October 2026 MTD (Oct 14, 3 days) + September 2026 (full month) + August 2026*
QC Status: *RECONCILED All numbers independently verified against raw AWS API data*

Executive Summary

Field	Value
Account	730649731674 (Himanshu Gohil)
Account Alias	None configured
Audit Date	2026-10-04
Audit Period	Oct 2026 MTD (Oct 14) + Sep 2026 (full)
Sep 2026 CE Total (net)	-$0.0000008495 (~$0.00)
Sep 2026 Gross Charges	$0.4342 (before credits)
Sep 2026 Credits Applied	-$0.4342
Oct 2026 CE Total (net, 3 days)	-$0.0000002541 (~$0.00)
Oct 2026 Gross Charges	$0.0328 (before credits)
Oct 2026 Credits Applied	-$0.0328
Total Resource Objects in Inventory	10
Resources with Current Net Cost	0 (all offset by credits or free tier)
Resources with Conditional Cost	3 (EBS volume + 2 KMS CMKs, if credits expire)
Active Cost Anomalies	0 (none current)
Historical Anomalies Detected	8 distinct events (all ended Sep 30, 2026)
Confirmed Current Net Savings	$0.00 (current net cost is already $0)
Conditional Savings (if credits expire)	$2.80/month ($0.80 EBS + $2.00 KMS)
Top Historical Cost Driver	Cross-region transfer ap-south-1ap-southeast-1: $0.3511 (Sep 2026)
Top Current Action	Delete orphaned EBS vol-008573e93270b1bcd (conditional savings $0.80/month)

Summary: This AWS account is used for lab/learning environments. All compute infrastructure (EC2, EKS, ECS, Lambda, RDS, NAT Gateways, Load Balancers) was fully decommissioned by September 30, 2026. Both the September and October net costs are effectively **$0.00** AWS credits fully offset all gross charges. No active anomalies exist. The account retains 10 passive resources (1 orphaned EBS volume, 2 S3 buckets, 5 CloudWatch log groups, 2 KMS CMKs). If AWS credits expire, the conditional exposure is $2.80/month from the orphaned EBS volume and KMS keys.

1. Cost Overview

All costs are AWS Cost Explorer UnblendedCost values, which equal BlendedCost and NetUnblendedCost for this account (no RI/SP amortization). Negative values represent credits/discounts applied by AWS.

Service	Sep 2026 Gross	Sep 2026 Net	Oct MTD Gross	Oct MTD Net	Notes
EC2 - Other	$0.4342	Offset	$0.0328	Offset	Largest line; includes data transfer, NAT, EBS
AWS Data Transfer (credit)		-$0.4342		-$0.0328	AWS credit fully offsets EC2-Other
Amazon EC2 Compute	$0.0000052	Offset	$0.0000069	Offset	Near-zero; free tier or credits
Amazon ELB	$0.0000025	Offset	$0.0000001	Offset	Near-zero residual
Amazon RDS	$0.0000002	Offset	$0.0000000	Offset	Near-zero residual
Amazon S3	$0.0000014	Offset	$0.0000000	Offset	2 small buckets
Amazon ECR	$0.0000000	Offset	$0.0000000	Offset	Near-zero
Amazon ECS	$0.0000000	Offset	$0.00		No activity
Amazon EKS	-$0.0000000	Offset	$0.00		No clusters
AWS KMS	-$0.0000008	-$0.0000008	-$0.0000003	-$0.0000003	Net negative (credit exceeds charge)
AWS Backup	-$0.0000000	-$0.0000000	-$0.0000000		Near-zero
AWS WAF	-$0.0000001	-$0.0000001	$0.00		Historical only
All Others	$0.00	$0.00	$0.00	$0.00	Free tier; Lambda, VPC, CW, etc.
CE Reported Total		**-$0.0000008**		**-$0.0000003**	**Effectively $0.00**

How to read this table: *"Gross" is the list-price charge before credits. "Net" is after AWS credits are applied. For this account, all positive gross charges are offset by negative AWS Data Transfer credits, resulting in a near-zero net total. Only the KMS and Backup lines remain slightly negative (over-credit).*

2. Top Cost Drivers

All costs below are HISTORICAL (September 2026 lab period). Current net cost for all items is $0.00.

Rank	Usage Type / Service	Quantity	AWS-Reported Cost (Sep)	Pricing Basis	Status
1	Cross-region transfer: ap-south-1 ap-southeast-1 (APS3-APS1-AWS-Out-Bytes)	4.082 GB	$0.3511	AWS-reported	HISTORICAL
2	Cross-AZ transfer within us-west-2 (USW2-DataTransfer-Regional-Bytes)	9.055 GB	$0.0831	AWS-reported	HISTORICAL
3	NAT Gateway hours, us-west-2 (USW2-NatGateway-Hours)	42 hrs	$0.00 net (list: $1.89)	Offset by credits	HISTORICAL

Rank	Usage Type / Service	Quantity	AWS-Reported Cost (Sep)	Pricing Basis	Status
4	NAT Gateway data, us-west-2 (USW2-NatGateway-Bytes)	7.652 GB	$0.00 net (list: $0.34)	Offset by credits	HISTORICAL
5	EBS gp3 volume, ap-south-1 (APS3-EBS:VolumeUsage.gp3)	2.005 GB-Month	$0.00 net	Offset by credits	HISTORICAL
6	USW2-USE1-AWS-Out-Bytes (us-west-2 us-east-1)	0.001 GB	$0.0000189	AWS-reported	HISTORICAL

Note on October EC2-Other: Oct MTD contains USW2-DataTransfer-Regional-Bytes: 4.278 GB $0.0328 gross (fully offset by credits net $0.00). This is a residual transfer, likely from the newly-created orphaned EBS volume (created Oct 1) generating API/metadata traffic. No active EKS or NAT Gateway is running in October.

3. Cost Anomalies

AWS Cost Anomaly Detection reported 8 distinct anomaly events. All are HISTORICAL ended on or before September 30, 2026. ***TotalActualSpend values in anomalies are per-event daily amounts, not additive monthly totals. They do not add to the CE monthly total; they are point-in-time spikes fully contained within the Sep 2026 service-level costs.***

Anomaly ID (short)	Date Range	Service	Actual Spend (event)	Expected	Impact	Root Cause	Status
62bfe040	Sep 2930	EBS / NAT (us-west-2)	$2.84	$0.07	+3,957%	NAT Gateway 42 hrs + EBS gp3 + NAT data 7.65 GB during EKS lab	HISTORICAL
19e656b1	Sep 2930	EC2 Compute	$4.03	$0.00	N/A	c7i-flex.large instances in us-west-2 and ap-south-1 (EKS nodes)	HISTORICAL
2ebc27d9	Sep 2930	Elastic Load Balancing	$1.64	$0.00	N/A	ALB/NLB from EKS Kubernetes LoadBalancer services	HISTORICAL
466b10ef	Sep 2930	Amazon VPC	$1.14	$0.00	N/A	NAT Gateway EIP, VPC flow logs, or interface endpoints	HISTORICAL
6367e764	Sep 2930	AWS KMS	$0.12	$0.06	+100%	Increased KMS API calls from active EKS encryption	HISTORICAL

Anomaly ID (short)	Date Range	Service	Actual Spend (event)	Expected	Impact	Root Cause	Status
e7a6aa0d	Sep 25	Amazon EBS	$0.44	$0.04	+1,000%	EBS volume activity during lab setup	HISTORICAL
acf4f31a	Sep 25	AWS WAF	$0.06	$0.00	N/A	WAF Web ACLactivated (aws-waf-logs-waf-lab-web-acl)	HISTORICAL
06a66db7	Sep 23	EC2 Compute	$0.55	$0.01	+5,400%	Early lab instance activity	HISTORICAL

Anomaly vs CE Reconciliation: *The sum of anomaly TotalActualSpend values ($11.96) is intentionally much larger than the CE monthly total (~$0.4342 gross). Anomaly Detection captures per-service spend during anomalous daily periods. These are not independent monthly charges they are sub-periods within the same September billing cycle. The $4.03 EC2 anomaly, $1.64 ELB anomaly, etc., are daily peaks; their underlying charges are captured in the EC2-Other ($0.4342) and other service lines in Cost Explorer and fully offset by credits.*

4. Active Resource Inventory

Resource count reconciliation: ***10 total objects*** *across all active regions.*

#	Category	Resource ID / Name	Region	State	Stored Data	AWS-Reported Net Cost	Conditional Cost
1	EBS Volume (gp3)	vol-008573e93270b1bcd	us-west-2a	**available (UNATTACHED)**	10 GB (created 2026-10-01)	$0.00 net (credit offset)	$0.80/month if credits expire
2	S3 Bucket	him-demo-18	us-east-1	Active (created 2026-09-07)	Unknown (metrics unavailable)	$0.00	Negligible
3	S3 Bucket	minishop-terraform-state-730649731674	us-east-1	Active (created 2026-09-22)	Unknown (metrics unavailable)	$0.00	Negligible
4	CW Log Group	/aws/ecs/containerinsights/incidentiq-target/performance	us-east-1	Active, 1-day retention	0 bytes	$0.00	$0.00
5	CW Log Group	/aws/eks/HimCluser/cluster	us-west-2	Active, **no retention**	6,309,180 bytes (6.02 MB)	$0.00 (free tier)	$0.00 (free tier)
6	CW Log Group	/aws/eks/himanshu-eks-cluster/cluster	ap-south-1	Active, **no retention**	32,346,158 bytes (30.85 MB)	$0.00 (free tier)	$0.00 (free tier)
7	CW Log Group	/aws/lambda/saa8-api-handler	ap-south-1	Active, **no retention**	13,162 bytes (0.01 MB)	$0.00 (free tier)	$0.00 (free tier)
8	CW Log Group	aws-waf-logs-waf-lab-web-acl	ap-south-1	Active, **no retention**	623,670 bytes (0.60 MB)	$0.00 (free tier)	$0.00 (free tier)

#	Category	Resource ID / Name	Region	State	Stored Data	AWS-Reported Net Cost	Conditional Cost
9	KMS CMK	18a08257-7685-4466-a27e-4b2870f552ae	us-east-1	Active (customer managed)	N/A	$0.00 net (credit offset)	$1.00/month if credits expire
10	KMS CMK	e1975ab0-8724-4335-8ae4-296958bca994	us-east-1	Active (customer managed)	N/A	$0.00 net (credit offset)	$1.00/month if credits expire

Resource count breakdown:

- Total resource objects: **10**
- Resources with current net cost > $0: **0**
- Resources with conditional cost (if credits expire): **3** (EBS + 2 KMS)
- Resources with zero cost now and in future: **5** (CloudWatch log groups within free tier)
- Resources with zero cost now, negligible future cost: **2** (S3 buckets)

CloudWatch total storage: 39,292,170 bytes = **37.48 MB = 0.0366 GB** entirely within the 5 GB free tier. **Current storage cost = $0.00.**

Resources confirmed NOT running:

- EC2 instances: 0 (all regions)
- EKS clusters: 0 (all regions)
- ECS clusters: 0 (all regions)
- Lambda functions: 0 (all regions)
- NAT Gateways: 0 (all regions)
- Load Balancers (ALB/NLB): 0 (all regions)
- RDS instances: 0 (all regions)
- EBS snapshots: 0 (all regions)
- AMIs (self-owned): 0
- ECR repositories: 0 (all regions)
- VPC endpoints: 0

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5. Optimization Opportunities

Priority	Finding	Current Net Cost	Conditional Monthly Savings	Classification	Confidence	Effort
P2	Delete orphaned EBS vol-008573e93270b1bcd (10 GB gp3, us-west-2)	$0.00 (credit offset)	$0.80/month if credits expire	Conditional	High	Low
P2	Set retention policies on CW log groups without retention (4 groups)	$0.00 (free tier)	$0.00 (preventive hygiene)	Preventive	High	Low
P2	Delete orphaned EKS log groups (clusters no longer exist)	$0.00 (free tier)	$0.00 (preventive hygiene)	Preventive	High	Low

Priority	Finding	Current Net Cost	Conditional Monthly Savings	Classification	Confidence	Effort
P3	Review and delete unused KMS CMKs (2 keys, us-east-1)	$0.00 (credit offset)	$2.00/month if credits expire	Conditional	Medium	LowMedium
P3	Add S3 lifecycle policies to both buckets	$0.00	$0.00 (preventive)	Preventive	Low	Low
P3	Consolidate future lab workloads to single region	N/A (historical)	$0.351/lab-session	Historical/Future	Medium	Medium
P3	Use S3 Gateway Endpoint in future labs (eliminate NAT-to-S3)	N/A (no NAT running)	Variable per lab	Historical/Future	High	Low

6. Detailed Findings

Finding 1: Orphaned EBS Volume Conditional Cost

Field	Value
Finding	Unattached EBS gp3 volume in us-west-2; no workload attached
Resource	vol-008573e93270b1bcd
Region / AZ	us-west-2a
Volume Type / Size	gp3, 10 GB
Volume Created	2026-10-01T04:57:09Z (created Oct 1, not before Sep 30)
State	available (UNATTACHED)
AWS-Reported Net Cost (Oct MTD)	-$0.0000000029 (credit offset net cost is $0.00)
AWS-Reported Usage (Oct 1–4)	2.7731 GB-Month (confirmed: 10 GB 3/31 days 0.968 GB-Month per day 3 days = 2.90 GB-Month; API value reflects partial-day precision)
Calculated Monthly Cost	10 GB $0.08/GB-month = **$0.80/month** (pricing assumption: AWS published gp3 price, us-west-2)
Savings Classification	CONDITIONAL $0.80/month IF credits expire
Confirmed Current Savings	$0.00 (credits currently absorb the charge)
Estimated Annual Savings (conditional)	$9.60/year
Confidence	High (volume confirmed unattached via describe-volumes API)
Why It Exists	EKS lab cluster was deleted but EBS volume was not included in teardown; created Oct 1 as part of final lab activity
Recommendation	Snapshot if data is needed; then delete
Risk	Data loss if deleted without snapshot (irreversible)

Field	Value
Human Approval Required	YES

Finding 2: CloudWatch Log Groups from Deleted EKS Clusters Preventive

Field	Value
Finding	Log groups from deleted EKS clusters persist with no retention policy
Resources	/aws/eks/HimCluser/cluster (us-west-2, 6.02 MB); /aws/eks/himanshu-eks-cluster/cluster (ap-south-1, 30.85 MB)
Total Stored	38,655,338 bytes = **36.88 MB = 0.0360 GB**
AWS-Reported Net Cost	$0.00 (all storage within 5 GB free tier)
Calculated Cost	0.0360 GB $0.03/GB-month = **$0.00108/month** (below free tier actual current cost = $0.00)
Free Tier Remaining	5 GB 0.0366 GB (all log groups combined) = **4.9634 GB free tier remaining**
Savings Classification	PREVENTIVE current cost is $0.00; prevents future accumulation
Why They Exist	AWS does not delete log groups when EKS clusters are deleted; these clusters no longer exist
EKS Clusters Confirmed Absent	Yes `eks list-clusters` returned empty in both us-west-2 and ap-south-1
Recommendation	Delete both orphaned EKS log groups; set 30-day retention on remaining groups
Risk	Log data permanently lost; no impact on running workloads (no clusters exist)
Human Approval Required	YES

Finding 3: CloudWatch Log Groups Missing Retention Policies

Field	Value
Finding	4 of 5 log groups have no retention policy; will accumulate logs indefinitely
Affected Groups	/aws/eks/HimCluser/cluster, /aws/eks/himanshu-eks-cluster/cluster, /aws/lambda/saa8-api-handler, aws-waf-logs-waf-lab-web-acl
Current Storage Cost	$0.00 (all within 5 GB free tier)
Risk	Without retention, storage grows without limit; cost begins at $0.03/GB-month beyond 5 GB
Confidence	High retentionInDays=null confirmed via describe-log-groups API
Recommendation	Set 30-day retention on active groups; delete the two orphaned EKS groups
Human Approval Required	YES setting retention will eventually delete old log data

Finding 4: KMS Customer Managed Keys Conditional Cost

Field	Value
Finding	2 CMKs exist in us-east-1; list price $1.00/month each
Key IDs	18a08257-7685-4466-a27e-4b2870f552ae; e1975ab0-8724-4335-8ae4-296958bca994
AWS-Reported Net Cost	$0.00 (KMS UnblendedCost = -$0.0000008404 in Sep; -$0.000000284 in Oct negative means credit exceeds charge)
List Price	$1.00/month per CMK (AWS published) = $2.00/month for 2 keys
Savings Classification	CONDITIONAL $2.00/month IF credits expire AND keys are confirmed unused
Confirmed Current Savings	$0.00 (current cost is $0.00 after credits)
Utilization	Utilization could not be determined from the available AWS data (CloudTrail not queried; key aliases/policies not retrieved)
Confidence	Medium (keys exist and are charged at list price before credits; necessity unverified)
Risk	HIGH deleting a CMK that encrypts live data renders that data permanently inaccessible
Human Approval Required	YES CMK deletion is irreversible after the 7-day waiting period

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Finding 5: Cross-Region Data Transfer ap-south-1 to ap-southeast-1 (HISTORICAL)

Field	Value
Finding	Cross-region data transfer from ap-south-1 to ap-southeast-1 during September 2026 lab
Usage Type	APS3-APS1-AWS-Out-Bytes
Quantity	4.0820 GB (AWS-reported)
AWS-Reported Cost	$0.3511 (gross; offset by credits net $0.00)
Calculated Cost	4.0820 GB $0.086/GB = $0.3511 (matches AWS-reported; price assumption confirmed)
Current Cost	$0.00 (no cross-region workloads running in October)
Historical Cost	$0.3511 (September 2026 lab session)
Root Cause	EKS workload in ap-south-1 communicating with service/registry in ap-southeast-1
Savings Classification	HISTORICAL/FUTURE no current cost; guidance for future labs
Estimated Savings per Lab Session	Up to $0.35 per session (frequency unknown not annualized)
Human Approval Required	N/A no current resources

Finding 6: Cross-AZ Data Transfer us-west-2 (HISTORICAL + RESIDUAL)

Field	Value
Finding	Cross-AZ traffic within us-west-2 historically from EKS; residual amount in Oct
Usage Type	USW2-DataTransfer-Regional-Bytes
Sep 2026 Quantity	9.055 GB

Field	Value
Sep 2026 AWS-Reported Cost	$0.0831 (gross; fully offset by credits net $0.00)
Implied Rate	$0.0831 / 9.055 GB = $0.0092/GB (NOTE: AWS rate is $0.01/GB per direction; the $0.0092 implied rate reflects actual billing not the "2 directions" formula)
Oct MTD Quantity	4.278 GB (Oct 14, 3 days)
Oct MTD AWS-Reported Cost	$0.0328 (gross; fully offset by credits net $0.00)
Oct Rate Implied	$0.0328 / 4.278 GB = $0.0077/GB (AWS-reported; use this not a formula)
Savings Classification	HISTORICAL for Sep; CONDITIONAL for Oct (if credits expire)
Oct Residual Explanation	October cross-AZ charges are residual: the EBS volume created Oct 1 and associated API/management traffic. No EKS cluster or NAT Gateway is running.
Human Approval Required	N/A no current EKS cluster

Finding 7: NAT Gateway Usage September 2026 (HISTORICAL)

Field	Value
Finding	NAT Gateway ran for 42 hours with 7.65 GB data in us-west-2 during September 2026 lab
NAT Hours	42 hrs (USW2-NatGateway-Hours)
NAT Data	7.652 GB (USW2-NatGateway-Bytes)
Calculated Hourly Cost	42 hrs $0.045/hr = $1.89 (pricing assumption: AWS published us-west-2)
Calculated Data Cost	7.652 GB $0.045/GB = $0.34
Calculated Total	$2.23 (list price estimate INFORMATIONAL ONLY)
AWS-Reported Cost	$0.00 net (fully offset by credits)
Current NAT Gateways Running	0 (confirmed via describe-nat-gateways in all regions)
Savings Classification	HISTORICAL no current cost
Savings per Future Lab Session	Up to $2.23/session list price (frequency unknown not annualized)
Human Approval Required	N/A

7. Architecture Cost Risks

Risk 1: Cross-Region Traffic Pattern

Observed (Sep 2026): 4.08 GB transferred from ap-south-1 to ap-southeast-1 at $0.086/GB = $0.3511.

Root cause: EKS cluster in ap-south-1 communicating with service or container registry in ap-southeast-1. Likely container image pulls from an ECR repository in the wrong region, or an application service endpoint hard-coded to ap-southeast-1.

Mitigation for future labs:

- Keep all resources in a single region
- Deploy ECR repositories in the same region as EKS clusters

- Review application configuration for cross-region API calls
- Use regional AWS service endpoints

Risk 2: Cross-AZ Traffic Pattern

Observed (Sep 2026): 9.055 GB within us-west-2 at AWS-reported $0.0831.

Root cause: EKS pod-to-pod communication across Availability Zones. Kubernetes distributes pods across AZs by default for resilience, but inter-AZ traffic incurs cost.

Mitigation for future labs:

- Pod Topology Spread Constraints with `topologyKey: topology.kubernetes.io/zone`
- Kubernetes EndpointSlice topology hints for automatic same-AZ preference
- For non-HA lab environments, single-AZ node group configurations

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Risk 3: NAT Gateway Without VPC Endpoints

Observed (Sep 2026): 42 hrs NAT Gateway operation + 7.65 GB data processed. List-price estimate: $2.23/session (fully offset by credits in Sep).

Root cause: EKS worker nodes in private subnets required NAT Gateway to reach AWS services (ECR, STS, S3). Without VPC Endpoints, all AWS API traffic routes through NAT.

Mitigation for future labs:

- S3 Gateway Endpoint (free) eliminates S3 traffic via NAT
- ECR and STS Interface VPC Endpoints (~$7.20/month each, but eliminate per-GB NAT charges for container operations)
- For short-lived labs (< 8 hours total), NAT cost is minimal; endpoints worthwhile for multi-day labs

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Risk 4: EBS Volume Orphaning

Observed: vol-008573e93270b1bcd created 2026-10-01, unattached as of 2026-10-04. Conditional cost: $0.80/month (list price; currently credit-offset).

Root cause: EBS volumes are not deleted when EKS node groups or EC2 instances are terminated unless `DeleteOnTermination=true` is explicitly set.

Mitigation:

- Set `DeleteOnTermination=true` on all EBS volumes in Terraform/IaC
- AWS Config rule `ec2-volume-inuse-check` to flag unattached volumes automatically
- Quarterly EBS audit: `describe-volumes --filters Name=status,Values=available`

Risk 5: CloudWatch Log Group Orphaning

Observed: 2 EKS log groups persist after cluster deletion; 4 of 5 groups have no retention policy. All within free tier currently.

Root cause: AWS does not delete CloudWatch log groups when source services are deleted. Log groups must be managed independently.

Mitigation:

- Include `aws_cloudwatch_log_group` with `retention_in_days` in all Terraform modules
- Add log group cleanup to lab teardown runbooks
- AWS Config rule to enforce retention policies

Risk 6: KMS CMK Lifecycle

Observed: 2 CMKs in us-east-1 with no confirmed active encryption use. Conditional cost $2.00/month (list; currently credit-offset).

Root cause: CMKs created for lab EKS encryption or RDS encryption are not deleted when workloads are decommissioned.

Mitigation:

- Tag CMKs at creation: `Project: eks-lab-2026-09`
- Include CMK deletion in lab teardown checklist (note: AWS enforces 7-day waiting period)
- Review CloudTrail for last CMK usage before scheduling deletion

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Risk 7: Multi-Region Lab Footprint

Observed: Resources deployed across 3 regions (us-east-1, us-west-2, ap-south-1) simultaneously, resulting in cross-region data transfer costs.

Mitigation:

- Designate ap-south-1 as primary lab region (closest to operator, already most-used)
- Use us-west-2 only when specifically testing us-west-2 architectures
- Clean all region-specific residual resources (log groups, EBS, S3) after each lab

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8. Savings Summary

*Savings are categorized by type. **Confirmed current savings = $0.00** because current net cost is already $0.00 after credits. All actionable savings are conditional on credit expiry.*

Opportunity	Savings Type	Monthly Savings	Annual Savings	Confidence	Notes
Delete EBS vol-008 573e93270b1bcd	Conditional (if credits expire)	$0.80	$9.60	High	List price $0.08/GB 10 GB; currently credit-offset
Review/delete KMS CMKs (2 keys)	Conditional (if credits expire + confirmed unused)	$2.00	$24.00	Medium	List price $1.00/key; utilization unverified
Set CW log retention policies	Preventive	$0.00	$0.00	High	All 37.48 MB within 5 GB free tier
Delete orphaned EKS log groups	Preventive	$0.00	$0.00	High	Prevents unbounded growth
S3 lifecycle policies	Preventive	$0.00	$0.00	Low	Buckets currently near-empty
Cross-region transfer (future labs)	Future Lab Guidance	$0.351/session	Not annualized	Medium	Per-session; lab frequency unknown
Cross-AZ traffic (future labs)	Future Lab Guidance	$0.083/session	Not annualized	Medium	Per-session; lab frequency unknown
NAT Gateway (future labs)	Future Lab Guidance	$2.23/session est.	Not annualized	Medium	List price; currently credit-offset
Conditional Total (if credits expire)		**$2.80/month**	**$33.60/year**		EBS + KMS only; no double-counting
Confirmed Current Savings		**$0.00**	**$0.00**		Net cost is already $0

> ***No double-counting:*** *EBS, KMS, and CloudWatch are distinct, independent resources. Cross-region, cross-AZ, and NAT savings are per-session estimates and are not included in the monthly total because lab session frequency is unknown. Per-session estimates are not annualized without a defensible usage assumption.*

9. Recommended Action Plan

Immediate (This Week)

Action	Resource	Savings Impact	Risk	Approval
Verify EBS volume data (is it needed?)	vol-008573e93270b1bcd	Pre-requisite	None	Not required
Create EBS snapshot if data needed	vol-008573e93270b1bcd	Pre-requisite	None	Not required
Delete orphaned EBS volume	vol-008573e93270b1bcd	Eliminates conditional $0.80/month exposure	Low (snapshot first)	**Required**
Set 30-day retention on log groups lacking policy	4 CW log groups	Preventive hygiene	Low	Recommended
Delete /aws/eks/HimCluser/cluster log group	us-west-2 CloudWatch	Preventive; cluster confirmed gone	None	**Required**
Delete /aws/eks/himanshu-eks-cluster/cluster log group	ap-south-1 CloudWatch	Preventive; cluster confirmed gone	None	**Required**

Before Next Lab Session

Action	Resource	Savings Impact	Notes
Identify purpose of both KMS CMKs	us-east-1 KMS	Conditional $2.00/month	Check CloudTrail for last use; inspect key aliases/policies
If unused, schedule CMK deletion	2 CMKs	Conditional $2.00/month	HIGH RISK verify no live encrypted data first
Add lifecycle policies to both S3 buckets	him-demo-18, minishop-terraform-state	Preventive	Include noncurrent version expiration + multipart abort
Audit minishop-terraform-state contents	S3	Preventive	Confirm project decommissioned; remove stale state files

Architecture Improvements (Future Labs)

Improvement	Expected Benefit	Effort
Consolidate all lab work to ap-south-1 (single region)	Eliminate cross-region charges ($0.351/session observed)	Low
Add S3 Gateway Endpoint before deploying EKS	Reduce NAT data charges; endpoint is free	Low
Set DeleteOnTermination=true on all EBS in IaC	Prevent future orphaned volumes	Low
Include retention_in_days in all Terraform log group resources	Prevent log accumulation and hygiene issues	Low

Improvement	Expected Benefit	Effort
Use Pod Topology Spread Constraints in EKS	Reduce cross-AZ data transfer	Medium
Include CMK teardown in lab runbook	Prevent KMS CMK accumulation	Low

Continuous FinOps

- Enable AWS Budgets alert at $5/month to catch unexpected charges
- Enable AWS Config rule `ec2-volume-inuse-check` to flag unattached EBS automatically
- Use existing Cost Anomaly Detection monitors review weekly
- Run quarterly resource audit: EBS available, CW log groups without retention, KMS CMK aliases
- Tag all lab resources at creation: `Environment: lab`, `Project: `, `CreatedBy:`

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10. Safety / Assumptions

Pricing Assumptions

Item	Assumed Price	Source	Verification Status
EBS gp3 storage (us-west-2)	$0.08/GB-month	AWS published price	Consistent with AWS-reported Oct usage: 2.7731 GB-Month $0.2219 calc vs $0.0000000029 net (credit offset)
CloudWatch Logs storage	$0.03/GB-month (after 5 GB free tier)	AWS published price	Not verified against billing (all storage within free tier = $0.00 actual)
KMS CMK	$1.00/month per key	AWS published price	Consistent with Sep CE KMS line showing gross ~$0.0000008 before credits
Cross-region transfer (ap-south-1 ap-southeast-1)	$0.086/GB	AWS published price	Verified: 4.082 GB $0.086 = $0.3511 = AWS-reported $0.3511
Cross-AZ transfer	$0.01/GB per direction	AWS published price	AWS-reported: $0.0831 for 9.055 GB = $0.0092/GB implied. Do NOT use "2 directions" formula. Use AWS-reported amount as authoritative.
NAT Gateway (us-west-2)	$0.045/hr + $0.045/GB	AWS published price	List-price estimate only; actual AWS-reported cost = $0.00 (credit-offset)

Cost Source Definitions

Term	Definition
AWS-Reported Cost	Exact value from AWS Cost Explorer API (`UnblendedCost`). Authoritative. Never replaced with a calculated estimate.
Calculated Estimate	quantity unit price. Used only where AWS does not report a per-resource cost breakdown.
Conditional Cost	List-price charge that is currently offset by credits; becomes real if credits expire.

Term	Definition
Preventive	Zero current cost; recommendation is hygiene/risk management.
Historical	Cost incurred in a past period; no current exposure.

Credits and Offsets

AWS Cost Explorer shows the following credit structure for this account:

- **September 2026:** EC2-Other gross = $0.4342; AWS Data Transfer credit = -$0.4342. Net CE total = -$0.0000008495 (effectively $0.00).
- **October 2026 MTD:** EC2-Other gross = $0.0328; AWS Data Transfer credit = -$0.0328. Net CE total = -$0.0000002541 (effectively $0.00).
- The nature of these credits (promotional, AWS Activate, free tier, or billing adjustment) cannot be determined from Cost Explorer API alone.
- All "conditional savings" figures assume these credits may expire. If credits continue indefinitely, no additional action is required.

Data Availability Limitations

- **KMS:** Key aliases, policies, and CloudTrail usage history were not retrieved. Key necessity cannot be confirmed without this data. Marked as "utilization could not be determined from the available AWS data."
- **S3 bucket sizes:** CloudWatch S3 metrics returned no data points (buckets likely below monitoring threshold or metrics not enabled). Exact bucket sizes unknown; treated as near-zero based on creation dates and $0.00 billing.
- **CloudTrail:** Not queried. Last-use dates for KMS keys and S3 access patterns are unavailable.
- **Other regions (eu-*, ca-central-1, sa-east-1, ap-northeast-*, ap-southeast-2):** Not individually inventoried for resources; all showed $0.00 in Cost Explorer no active resources expected.

Human Approval Requirements

All remediation actions require explicit human approval. This report is advisory only. No AWS resources were created, modified, or deleted during this audit.

Actions requiring approval:

- Delete EBS volume vol-008573e93270b1bcd (irreversible without prior snapshot)
- Delete CloudWatch log groups (irreversible)
- Set CloudWatch log group retention policies (eventual data deletion)
- Schedule KMS CMK deletion (HIGH RISK irreversible after waiting period; verify first)
- Any S3 bucket or object modifications

FinOps Reconciliation & Quality Control

Cost Reconciliation

Reconciliation Check	Sep 2026	Oct MTD	Status
CE Total (no grouping, authoritative)	-$0.0000008495	-$0.0000002541	AWS-reported
Sum of service line items	-$0.0000008495	-$0.0000002541	Calculated
Gross charges (positive services only)	+$0.4341929690	+$0.0327901752	Calculated
Credits / discounts (negative services)	-$0.4341938185	-$0.0327904293	Calculated

Reconciliation Check	Sep 2026	Oct MTD	Status
Net (gross + credits)	-$0.0000008495	-$0.0000002541	Calculated
Reconciliation difference	**$0.0000000000**	**$0.0000000000**	**PASS**

Both months reconcile exactly to $0.00 difference between the CE total (ungrouped) and the sum of all service-level line items.

Calculation Validation

Finding	Quantity	Unit	Unit Price	Calculated Cost	AWS-Reported Cost	Match?
Cross-region transfer (APS3APS1)	4.0820	GB	$0.086/GB	$0.3511	$0.3511	YES
Cross-AZ transfer USW2 (Sep)	9.0549	GB	$0.0092/GB (implied)	$0.0831	$0.0831	YES (use AWS-reported)
EBS gp3 unattached (full month)	10	GB	$0.08/GB-month	$0.80/month	N/A (credit-offset)	Estimate only
NAT Gateway hours (Sep)	42	hrs	$0.045/hr	$1.89	$0.00 (credit-offset)	List-price estimate
NAT Gateway data (Sep)	7.652	GB	$0.045/GB	$0.34	$0.00 (credit-offset)	List-price estimate
EBS gp3 Oct MTD (3 days, 2.7731 GB-Mo)	2.7731	GB-Month	$0.08/GB-month	$0.2219	$0.00 (credit-offset)	List-price estimate
CW log storage (all groups)	0.0366	GB	$0.03/GB-month	$0.00 (free tier)	$0.00	YES
KMS CMK 2	2	keys	$1.00/key/month	$2.00/month	$0.00 (credit-offset)	List-price estimate

Savings Double-Counting Check

Saving	Source Resource	Independent?	Included in Monthly Total?
EBS volume deletion ($0.80/month conditional)	vol-008573e93270b1bcd	Yes	Yes
KMS CMK deletion ($2.00/month conditional)	2 separate CMKs	Yes independent of EBS	Yes
CW log retention ($0.00)	5 log groups	Yes independent of above	Yes ($0.00)
Cross-region savings ($0.351/session)	Historical lab behavior	Not monthly per-session	**No — excluded from monthly total**
Cross-AZ savings ($0.083/session)	Historical lab behavior	Not monthly per-session	**No — excluded from monthly total**
NAT Gateway savings ($2.23/session)	Historical lab behavior	Not monthly per-session	**No — excluded from monthly total**
Conditional monthly total			**$2.80/month (EBS + KMS only)**

Per-session historical savings ($0.351 + $0.083 + $2.23 = $2.66/session) are not included in the monthly total because lab session frequency is unknown. They cannot be annualized without a defensible assumption.

Resource Count Reconciliation

Category	Count	Current Net Cost	Classification
EBS volumes	1	$0.00 (credit-offset)	Conditional
S3 buckets	2	$0.00	Zero cost
CloudWatch log groups	5	$0.00 (free tier)	Zero cost
KMS CMKs	2	$0.00 (credit-offset)	Conditional
Total resource objects	**10**	**$0.00**	

*Prior report stated "6 active resources" **CORRECTED TO 10**. The prior count omitted 2 S3 buckets and 2 KMS CMKs, and miscounted the 5 CloudWatch log groups.*

Current vs Historical Validation

Cost Item	Classification	Justification
Sep 2026 EC2-Other ($0.4342 gross)	HISTORICAL	All Sep charges ended by Oct 1; no EC2/EKS/NAT running in Oct
Sep 2026 anomalies (all 8 events)	HISTORICAL	All anomaly end dates Sep 30, 2026; confirmed via API
Oct MTD EC2-Other ($0.0328 gross)	CURRENT (residual)	Cross-AZ data transfer tied to EBS volume created Oct 1 + API traffic; fully credit-offset
EBS volume $0.80/month list price	CONDITIONAL	Volume exists and is billable; currently credit-offset
KMS CMKs $2.00/month list price	CONDITIONAL	CMKs exist and carry list-price charges; currently credit-offset
CW log groups $0.00	CURRENT (zero)	All within 5 GB free tier; confirmed via storedBytes API values

Anomaly vs CE Total Reconciliation

Item	Value	Explanation
Sum of anomaly TotalActualSpend values	$11.96	Per-event daily spend during anomalous periods NOT additive monthly charges
Sep 2026 CE gross (full month)	$0.4342	Full-month aggregated charges across all service lines
Why anomaly sum > CE gross	N/A	Anomaly amounts represent gross service spend on specific days. The underlying EC2, ELB, VPC, EBS charges on those days ARE included within the $0.4342 EC2-Other, but the anomaly TotalActualSpend values include charges that are offset by credits the CE line items show net or credit-adjusted values. These amounts are NOT double-counted.

AWS-Reported vs Estimated Cost Validation

Finding	AWS-Reported (authoritative)	Calculated Estimate	Action
Cross-region transfer	$0.3511	$0.3511 (4.082 GB $0.086)	Estimates match; use AWS-reported
Cross-AZ transfer (Sep)	$0.0831	$0.0905 (9.055 GB $0.01)	Estimates differ; **use AWS-reported $0.0831**
EBS monthly cost	Not reported (credit-offset)	$0.80/month (10 GB $0.08)	AWS reports $0.00 net; estimate is conditional exposure
CW log storage	Not reported (free tier)	$0.0011/month (0.0366 GB $0.03)	Both zero (free tier); consistent
NAT Gateway	$0.00 net (credit-offset)	$2.23 list price	AWS reports $0.00; estimate is conditional/informational

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Report generated by Kiro FinOps Agent on 2026-10-04.
QC pass completed 2026-10-04. All numbers independently verified against raw AWS API data.
Access level: READ-ONLY. No AWS resources were modified during this audit.
All remediation actions require explicit human approval.